

Assignment 16 (ELEC 341 L16_FrequencyResponse)

Problem 1:

Convert the given transfer function into frequency response function, i.e., into $G(j\omega)$ form. Also, compute $M(\omega) = |G(j\omega)|$ and $\phi(\omega) = \angle G(j\omega)$. Find $20\log M(\omega)$ and $\phi(\omega)$ at $\omega = 20$.

$$G(s) = \frac{(s+3)(s+5)}{s(s+2)(s+4)} = \frac{s^2 + 8s + 15}{s^3 + 6s^2 + 8s}$$

$$G(j\omega) = \frac{(j\omega)^2 + 8j\omega + 15}{(j\omega)^3 + 6(j\omega)^2 + 8j\omega} \quad \begin{array}{l} j^3 = j \cdot j^2 \\ = -j \end{array}$$

$$= \frac{-\omega^2 + 8j\omega + 15}{-j\omega^3 - 6\omega^2 + 8j\omega} = \frac{15 - \omega^2 + 8j\omega}{-6\omega^2 + j(8\omega - \omega^3)}$$

$$G(j\omega) = \frac{15 - \omega^2 + 8j\omega}{-6\omega^2 + j(8\omega - \omega^3)}$$

$$M(\omega) = |G(j\omega)| = \frac{\sqrt{(15 - \omega^2)^2 + (8\omega)^2}}{\sqrt{(6\omega^2)^2 + (8\omega - \omega^3)^2}}$$

$$M(20) = \frac{\sqrt{(15 - 20^2)^2 + 8(20)^2}}{\sqrt{(6(20)^2)^2 + (8(20) - 20^3)^2}} \approx 0.0508497445458$$

$$20\log(M(20)) = -25.8742 \text{ dB}$$

$$\phi(\omega) = \angle G(j\omega)$$

$$= \tan^{-1}\left(\frac{8\omega}{15 - \omega^2}\right) - \tan^{-1}\left(\frac{8\omega - \omega^3}{-6\omega^2}\right)$$

$$\phi(\omega) = \tan^{-1}\left(\frac{8\omega}{15 - \omega^2}\right) - \tan^{-1}\left(\frac{8\omega - \omega^3}{-6\omega^2}\right)$$

$$\phi(20) = -95.5465^\circ$$